TIC 314239566

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       2.572641 d
  Epoch (t0)   1599.1559 BTJD
  Duration     0.1700 d
  Depth        2.062 %  (20,620 ppm)
  SNR          148.4

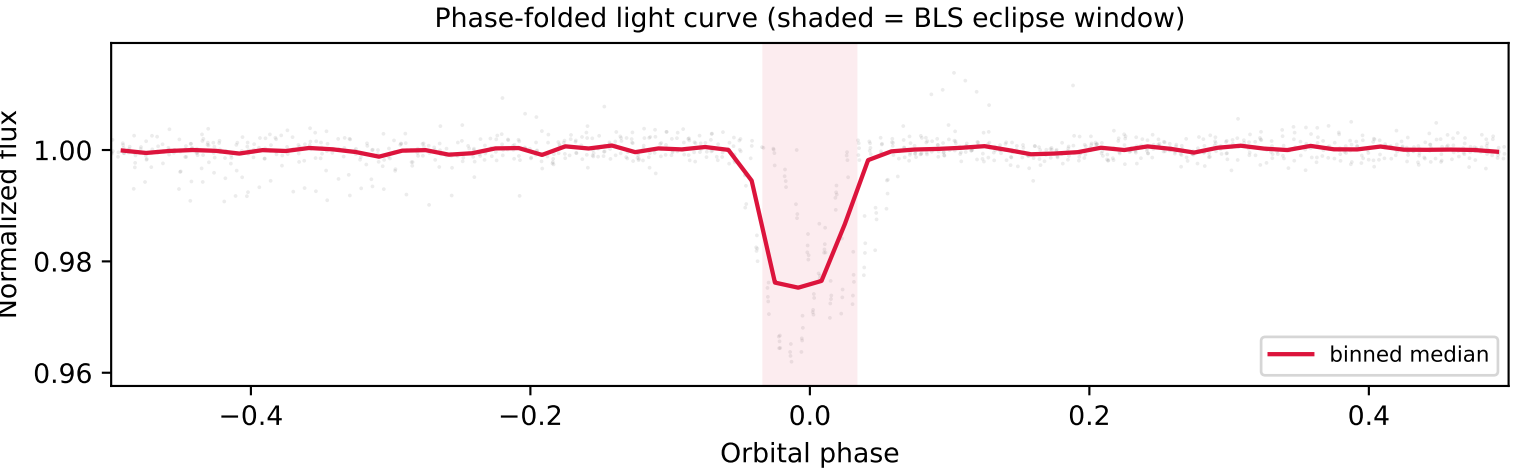
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     10  (gate >= 3)
  Robust depth ratio    1.09  (gate >= 0.50)
  Robust depth sigma    96.9  (gate >= 7.0)
  Odd/even z-score      1.18  (consistent)
  Secondary eclipse     not detected
```

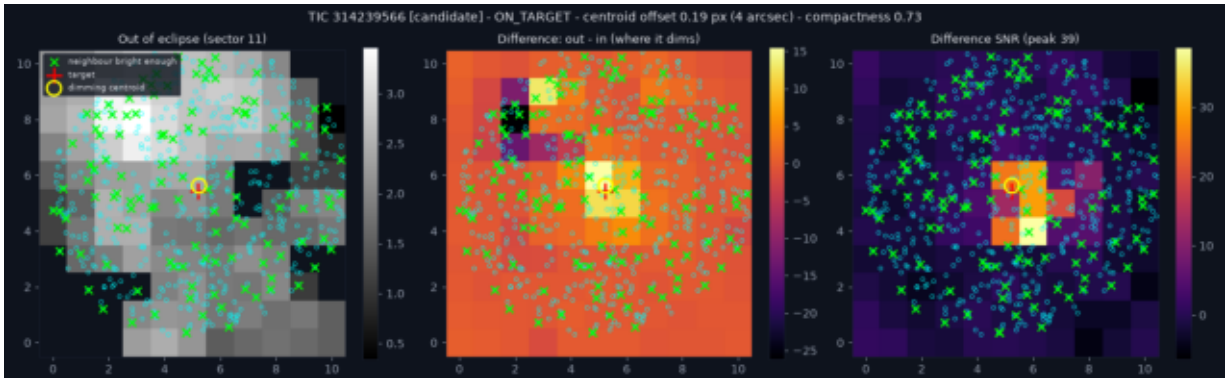
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 90.5 sigma    t0 refined by -0.0075 phase
  Difference-image peak SNR   38.65   (gate >= 5.0)      Compactness   0.73   (gate >= 0.50)
  Dimming centroid offset     0.19 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 314239565
    0.34 px away, dTmag +3.74 - farther from the dimming than the target
```





No caveats - clean detection on every check.